

Residual-Calibrated Multivariate Forecasting and Stress Testing under Distributional Change

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August 30, 2026

Abstract

Reliable financial forecasts require more than accurate conditional means. They also require predictive distributions that represent joint uncertainty, tail risk, and the possibility that market behavior changes after a model is fitted. This study develops an interpretable framework for calibrated and robust multivariate financial forecasting. A vector autoregression (VAR) captures predictable interactions among a small set of related asset returns, while competing joint residual models represent the uncertainty left unexplained by the VAR. Gaussian, multivariate Student- t , empirical bootstrap, and diffusion-based innovation specifications are compared within the same conditional-mean model. Sampled innovations are propagated recursively to construct probabilistic return, price, and portfolio paths.

The framework evaluates whether forecast probabilities are credible before treating flexibility as evidence of robustness. Marginal, multivariate, and tail calibration are assessed alongside proper scoring rules and sharpness. Innovation-based stress tests then alter volatility, dependence, downside tails, and contamination to measure how forecast calibration and portfolio-risk estimates deteriorate under distributional change. Controlled simulations establish behavior under known innovation laws, while a rolling financial case study evaluates external validity. Diffusion-Innovation VAR (DI-VAR) is treated as an unconditional generative baseline. The proposed Conditional Diffusion-Innovation VAR (CDI-VAR) instead models standardized joint innovations conditionally on recent residual history and a causal volatility state, then applies bounded, regularized, and adaptively updated calibration to completed VAR forecast distributions. The objective is to determine which residual models produce reliable forecasts, under which conditions, and at what cost.

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1 Introduction

Multivariate financial forecasting has two distinct tasks. The first is to estimate which components of future asset behavior are predictable from recent cross-asset dynamics. The second is to characterize the uncertainty that remains after those dynamics have been modeled. Point forecasts address only the first task; risk management, interval estimation, and portfolio decisions require the second.

Vector autoregressions provide a transparent model of lagged interactions among financial variables. Their coefficient matrices can be inspected, stability can be diagnosed, and forecast paths can be generated recursively. Yet the usual Gaussian innovation assumption is difficult to reconcile with financial returns that exhibit heavy tails, asymmetric losses, volatility clustering, nonlinear dependence, and stronger co-movement during market stress. A VAR may therefore provide an adequate conditional-mean forecast while producing an unreliable predictive distribution.

This study separates conditional-mean modeling from uncertainty modeling. The VAR captures predictable cross-asset structure, and its rolling forecast errors represent what the model failed to explain. Alternative joint innovation models are fitted to those errors and propagated through the same VAR. This controlled comparison asks whether additional distributional flexibility produces forecasts that are not only sharper, but also better calibrated on chronologically later observations.

Diffusion models are included as a flexible method for learning joint residual distributions. The resulting Diffusion-Innovation VAR (DI-VAR) does not replace the VAR or directly model prices; it replaces the conventional parametric innovation law with samples from a learned denoising process. DI-VAR is compared with simpler Gaussian, Student- t , and empirical bootstrap alternatives. This hierarchy is essential because low-dimensional financial residuals may not contain enough information to justify a complex generative model.

This thesis also develops Conditional Diffusion-Innovation VAR (CDI-VAR). CDI-VAR retains the transparent VAR conditional mean but addresses two limitations of unconditional residual generation: volatility clustering and state-dependent residual behavior. It causally standardizes rolling VAR errors, conditions a joint diffusion law on recent standardized errors and current volatility, and recalibrates final forecast deviations using only previously observable forecast outcomes. Calibration is deliberately separated from state evolution so a widening correction cannot recursively manufacture higher future volatility.

Robustness is distinguished from flexibility. A model that closely reproduces historical residuals may still fail when volatility, dependence, or tail behavior changes. The framework therefore evaluates nominal calibration first and subsequently exposes every innovation model to the same controlled stresses. Robustness is measured by the rate at which calibration, proper scores, and portfolio-risk estimates deteriorate as the forecast environment moves away from its reference distribution.

The central research question is:

How does the choice of joint residual model affect the calibration and robustness of multivariate autoregressive forecasts under changes in volatility, dependence, and tail behavior?

The methodological sub-question is:

Under what residual structures and distributional shifts does diffusion-based innovation

modeling provide meaningful improvements over Gaussian, Student-t, and empirical alternatives?

The contribution is a modular methodology connecting interpretable dynamics, joint uncertainty modeling, calibration, and distributional stress. The study remains informative if a simpler innovation model is preferable.

2 Literature Review

Stub for thesis development. This section will position the work relative to: multivariate autoregressive forecasting; probabilistic forecast calibration and proper scoring rules; conditional heteroskedasticity and volatility standardization; residual and block bootstrap methods; diffusion and score-based generative models for time series; online and conformal calibration; and distributionally robust financial forecasting. The final review will distinguish unconditional generative innovation models from conditional and adaptively calibrated predictive distributions, and will state precisely which methodological gap CDI-VAR is intended to address. Citations and claims will be added only after a systematic source review.

3 Conceptual Framework

The methodology contains four layers:

1. **Predictable dynamics.** A VAR estimates lagged interactions among related asset returns.
2. **Joint residual uncertainty.** Competing innovation models represent multivariate behavior left unexplained by the VAR.
3. **Calibration.** Forecast distributions are tested for marginal, joint, and tail reliability together with sharpness.
4. **Stress and robustness.** Controlled changes to the innovation law reveal how statistical credibility and portfolio-risk estimates deteriorate.

This decomposition gives each component an interpretable function. VAR coefficients describe mean transmission across assets. Residual diagnostics identify shape and dependence omitted by the mean model. Innovation-model comparisons determine whether extra distributional complexity is useful. Calibration determines whether reported probabilities correspond to observed frequencies. Stress-response curves identify failure thresholds and translate them into financial consequences.

4 Statistical Methodology

4.1 Financial representation

Let $P_t \in \mathbb{R}_+^K$ denote adjusted prices for K related assets. The default modeled variable is the vector of log returns,

$$y_t = \log P_t - \log P_{t-1}, \tag{1}$$

where the logarithm is applied componentwise. Returns are preferred to raw prices because a standard VAR requires approximately stationary inputs. If long-run equilibrium relationships among price levels are substantively important, a vector error-correction model is a separate extension.

The analysis uses a chronological training, calibration, and test split. The training period estimates conditional-mean dynamics. The calibration period supplies pseudo-out-of-sample forecast errors and model-selection information. The final test period remains untouched until the forecasting procedure is fixed.

4.2 Vector autoregressive dynamics

For $y_t \in \mathbb{R}^K$, a VAR of order p is

$$y_t = c + \sum_{\ell=1}^p A_\ell y_{t-\ell} + u_t, \quad (2)$$

where $c \in \mathbb{R}^K$, $A_\ell \in \mathbb{R}^{K \times K}$, and $u_t \in \mathbb{R}^K$ is the innovation vector. Lag order is selected using training or calibration observations, and the fitted system is checked for stability.

The VAR is intentionally common across innovation specifications. Holding the conditional mean fixed isolates the effect of uncertainty modeling. More complex forecasting systems can be considered in later robustness checks but are not required for the primary question.

4.3 Rolling forecast residuals

Purely in-sample residuals can understate deployment error. Innovation models are therefore trained on rolling one-step-ahead errors. At each calibration time t , the VAR is fitted or updated using information available through $t - 1$, and

$$\hat{u}_t = y_t - \hat{y}_{t|t-1} \quad (3)$$

is recorded. Each residual remains a joint K -dimensional observation so contemporaneous dependence is preserved.

If pronounced conditional heteroskedasticity remains, an auxiliary scale model may decompose

$$u_t = D_t z_t, \quad (4)$$

where D_t contains time-varying marginal scales and the innovation model learns standardized shocks z_t . Reporting results with and without standardization distinguishes flexible distribution learning from volatility forecasting.

4.4 Competing joint innovation models

Gaussian-VAR. The conventional benchmark assumes

$$u_t \sim \mathcal{N}(0, \hat{\Sigma}). \quad (5)$$

It is interpretable and efficient when the Gaussian assumption is adequate.

Student- t -VAR. A multivariate Student- t distribution provides a parsimonious heavy-tail model,

$$u_t \sim t_\nu(0, \hat{\Sigma}). \quad (6)$$

Its degrees-of-freedom parameter provides an interpretable measure of tail thickness.

Bootstrap-VAR. Joint residual vectors are sampled from the empirical residual distribution. A block-bootstrap variant preserves short-range temporal dependence. This is a strong low-dimensional nonparametric baseline.

Diffusion-Innovation VAR. DI-VAR fits a denoising diffusion model to joint residual vectors. For a clean residual u_0 , the forward process is

$$u_s = \sqrt{\bar{\alpha}_s} u_0 + \sqrt{1 - \bar{\alpha}_s} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

A network $\epsilon_\theta(u_s, s)$ is trained through

$$\mathcal{L}(\theta) = \mathbb{E}_{u_0, s, \epsilon} \left[\|\epsilon - \epsilon_\theta(u_s, s)\|_2^2 \right]. \quad (8)$$

The learned reverse process generates joint innovations without imposing a Gaussian or elliptical form. Because the target dimension is small, model capacity is deliberately limited and performance is reported relative to computational cost and sample size.

4.5 Conditional Diffusion-Innovation VAR

CDI-VAR extends DI-VAR by separating each innovation into a predictable scale and a conditionally generated standardized shock. Let $\hat{u}_t \in \mathbb{R}^K$ be a rolling one-step VAR forecast error. For component j , define

$$v_{t,j} = \lambda v_{t-1,j} + (1 - \lambda) \hat{u}_{t-1,j}^2, \quad (9)$$

$$\hat{z}_{t,j} = \frac{\hat{u}_{t,j}}{\sqrt{\max(v_{t,j}, \delta^2)}}, \quad (10)$$

where $\lambda \in (0, 1)$ is determined by the EWMA span and $\delta > 0$ is a numerical floor. The indexing in (9) is essential: v_t depends only on residuals observed before t , so standardization does not leak the magnitude of $\hat{u}_t$ into its own scale.

For residual-context length L , let

$$q_t = \left(\text{vec}(\hat{z}_{t-L:t-1}), \frac{\log \sqrt{v_t} - \mu_{\log \sigma}}{s_{\log \sigma}} \right) \in \mathbb{R}^{LK+K}, \quad (11)$$

where $\mu_{\log \sigma}$ and $s_{\log \sigma}$ are estimated from training observations only. Thus q_t contains recent standardized shock directions and the current marginal volatility regime while remaining observable before $\hat{u}_t$ occurs.

Conditional diffusion is fitted to $z_0 = \hat{z}_t$. At diffusion index s ,

$$z_s = \sqrt{\bar{\alpha}_s} z_0 + \sqrt{1 - \bar{\alpha}_s} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I_K), \quad (12)$$

and the conditional denoising network minimizes

$$\mathcal{L}_{\text{CDI}}(\theta) = \mathbb{E} \left[\|\epsilon - \text{epsilon}_\theta(z_s, s, q_t)\|_2^2 \right]. \quad (13)$$

DI-VAR estimates an unconditional law $p_\theta(u)$, whereas CDI-VAR estimates

$$p_\theta(z_t \mid z_{t-L:t-1}, v_t). \quad (14)$$

For simulation path m , a raw standardized draw $\tilde{z}_{t+h}^{(m)}$ is transformed into

$$\tilde{u}_{t+h}^{(m)} = \text{diag}\left(\sqrt{v_{t+h}^{(m)}}\right) \tilde{z}_{t+h}^{(m)}, \quad (15)$$

and the latent volatility state evolves according to

$$v_{t+h+1,j}^{(m)} = \lambda v_{t+h,j}^{(m)} + (1 - \lambda) \left(\tilde{u}_{t+h,j}^{(m)}\right)^2. \quad (16)$$

The raw standardized draw is appended to the residual context. This recursion allows volatility clustering and conditional joint shock geometry to affect uncertainty without replacing the interpretable VAR mean.

4.5.1 Chronological estimation

Rolling residuals are partitioned into three non-overlapping chronological blocks. The earliest block trains θ ; the next block selects the checkpoint by validation denoising loss; and the final block estimates forecast calibration. Test observations are excluded from all three operations.

4.5.2 Forecast-level scale calibration

Calibration is fitted to complete VAR forecasts rather than isolated residuals. At reserved calibration origin o , the VAR is refitted using observations strictly before o , raw CDI-VAR paths $Y_{o,h}^{(m)}$ are generated, and realized y_{o+h} is recorded. For

$$\mathcal{T}_c(x) = \tilde{x} + c(x - \tilde{x}), \quad (17)$$

where $\tilde{x}$ is the Monte Carlo median, the raw multiplier for nominal coverage $1 - \alpha$ is

$$c_h^{\text{raw}} \in \arg \min_{c \in \mathcal{G}} \left| \widehat{\text{Cov}}_h(c) - (1 - \alpha) \right|. \quad (18)$$

Here $\mathcal{G}$ is a fixed grid and $\widehat{\text{Cov}}_h(c)$ is empirical marginal coverage at lead h . Ties favor the multiplier closest to one.

Small calibration samples and regime changes can make (18) unstable. The deployed anchor is therefore

$$c_h^{\text{reg}} = \Pi_{[a,b]}(1 + \rho(c_h^{\text{raw}} - 1)), \quad (19)$$

where $\rho \in [0, 1]$ is shrinkage strength, $[a, b]$ contains one, and Π denotes projection onto that interval.

4.5.3 Online adaptive calibration

At rolling test origin o , each anchor uses at most the W most recent completed cases

$$\mathcal{A}_{o,h} = \{r < o : r + h \leq o\}. \quad (20)$$

The condition $r + h \leq o$ prevents look-ahead bias even for overlapping horizons. The raw online estimate from these cases is transformed by (19); before any outcome is available, the reserved-calibration estimate is the fallback.

For anchor set $\mathcal{H} = \{1, 5, 20\}$, the most recent lower anchor is carried forward between anchors. Final paths are

$$Y_{o,h}^{(m),\text{cal}} = \tilde{Y}_{o,h} + c_{o,h} \left(Y_{o,h}^{(m)} - \tilde{Y}_{o,h} \right). \quad (21)$$

The median is unchanged. Crucially, (21) is applied only after raw innovations have passed through the VAR. Calibrated deviations are never inserted into (11) or (16); this no-feedback condition prevents calibration from mechanically inflating subsequent volatility.

4.5.4 Mathematical justification

The construction has four motivations. Causal scale standardization reduces nonstationarity in the diffusion target while leaving volatility forecastable through an explicit state. Conditioning permits joint shock behavior to change with recent residual direction and volatility. Forecast-level calibration targets the predictive object actually evaluated rather than assuming residual calibration implies multi-step calibration. Finally, shrinkage and bounded online adaptation trade responsiveness against finite-sample variance while preserving the forecasting filtration. These choices do not guarantee calibration under arbitrary distribution shift; they define an auditable mechanism whose success can be measured at repeated origins.

4.6 Recursive probabilistic forecasts

For innovation model j , draw M future innovation paths. Each is propagated recursively through the fitted VAR:

$$y_{t+h}^{(m,j)} = \hat{c} + \sum_{\ell=1}^p \hat{A}_\ell y_{t+h-\ell}^{(m,j)} + \tilde{u}_{t+h}^{(m,j)}. \quad (22)$$

Return paths can be mapped back to prices through

$$P_{t+h}^{(m,j)} = P_t \odot \exp \left(\sum_{r=1}^h y_{t+r}^{(m,j)} \right), \quad (23)$$

where $\odot$ denotes componentwise multiplication. Because y_t contains asset log returns, portfolio arithmetic is performed after conversion to simple returns. For weights w satisfying $\sum_j w_j = 1$,

$$R_{t,j}^{(m)} = \exp(y_{t,j}^{(m)}) - 1, \quad (24)$$

$$R_{p,t}^{(m)} = \sum_{j=1}^K w_j R_{t,j}^{(m)}, \quad (25)$$

$$W_{p,t+h}^{(m)} = W_{p,t} \prod_{r=1}^h \left(1 + R_{p,t+r}^{(m)} \right). \quad (26)$$

This avoids the generally incorrect operation of weighting log returns and treating the result as an exact portfolio return.

5 Calibration and Forecast Evaluation

Calibration requires predicted probabilities to agree with observed frequencies across repeated forecast origins. It is evaluated with sharpness because arbitrarily wide intervals can attain high coverage without being useful.

5.1 Marginal evaluation

For every asset and forecast horizon, the study reports prediction-interval coverage, width, interval score, quantile loss, probability integral transform diagnostics, expected calibration error, and marginal Continuous Ranked Probability Score. Horizon-specific results are retained rather than reporting only pooled averages.

5.2 Joint evaluation

Marginal calibration does not imply that simultaneous outcomes are represented correctly. Joint evaluation therefore includes the Energy Score, dependence recovery, multivariate rank diagnostics where practical, and frequencies of simultaneous downside events. These quantities target the cross-sectional structure that joint innovation modeling is intended to preserve.

5.3 Tail and portfolio evaluation

Financial usefulness is assessed through lower-tail coverage, Value at Risk exceedance rates, Expected Shortfall, portfolio-loss distributions, and maximum drawdown. An innovation model can appear satisfactory at the asset level while materially understating joint portfolio losses.

5.4 Comparative protocol

Each innovation method uses the same VAR fits, forecast origins, horizons, and Monte Carlo sample counts. Results include mean performance, uncertainty across origins and random seeds, computational cost, and failure cases. DI-VAR and CDI-VAR are considered useful only when calibration, sharpness, or robustness gains are material relative to simpler alternatives.

6 Robustness and Distributional Stress

Robustness is the ability to retain useful probabilistic performance when the deployment innovation law differs from the reference law. The nominal model estimates a reference distribution $\hat{P}$ from calibration residuals. Stress operators generate a family Q_γ indexed by severity γ .

6.1 Stress scenarios

Volatility stress. Innovation scale is inflated according to

$$u^{\text{stress}} = \gamma u, \quad \gamma > 1. \quad (27)$$

Dependence stress. The innovation dependence structure is modified to increase common movement or downside correlation while controlling marginal scales. This tests diversification failure during market declines.

Tail stress. The frequency or magnitude of adverse innovations is increased through tail reweighting, contamination, or asymmetric transformations.

Asset-specific and outlier stress. A shock is applied to one component and propagated through the fitted system, or a controlled fraction of extreme joint residuals is introduced.

Historical stress. Residual patterns from turbulent periods are used as interpretable scenarios, subject to the limitations of small historical samples.

Wasserstein stress. Distributional ambiguity is summarized by

$$\mathcal{Q}_\varepsilon = \left\{ Q : W(Q, \hat{P}) \leq \varepsilon \right\}. \quad (28)$$

For a financial or calibration loss L , a robust risk summary is

$$\sup_{Q \in \mathcal{Q}_\varepsilon} \mathbb{E}_Q[L(Y)]. \quad (29)$$

The loss may represent portfolio loss, drawdown, tail exceedance, or calibration error. The Wasserstein radius is reported as a sensitivity parameter rather than assigned an unsupported universal interpretation.

6.2 Robustness curves

For each model and stress family, degradation relative to nominal performance is

$$\Delta_{m,j}(\gamma) = m_j(Q_\gamma) - m_j(\hat{P}), \quad (30)$$

where m is an error or loss metric. Robustness curves reveal whether a model fails gradually, abruptly, or only under particular shifts. This is more informative than declaring a model robust from nominal results.

7 Empirical Design

7.1 Financial case study

The primary application uses approximately four liquid and economically related assets. Asset selection is justified before comparative results are observed. Adjusted prices are aligned to common trading dates and converted to daily or weekly log returns. The analysis records missing-data decisions, corporate-action adjustments, sampling frequency, and portfolio weights.

Evaluation uses rolling forecast origins. At each origin, model fitting and innovation estimation use only prior observations. Hyperparameters are selected using training and calibration data, after which forecasts are evaluated on chronologically later outcomes. Multiple market regimes and test windows are included where sample size permits.

7.2 Implemented CDI-VAR configuration

The frozen primary configuration uses a VAR(1), $L = 5$ conditional residual lags, a 60-observation EWMA span, 50 diffusion steps, a hidden dimension of 128, and at most 300 epochs with chronological early stopping. Within the rolling residual sample, 15 percent is reserved for checkpoint validation and a later 15 percent for calibration. Calibration anchors are $\{1, 5, 20\}$ trading days with 32 Monte Carlo paths per reserved calibration origin. Raw multipliers are regularized with $\rho = 0.5$, projected to $[0.8, 1.25]$, and updated from at most $W = 12$ recently completed forecast cases. The primary evaluation uses 250 paths at 47 rolling origins separated by 20 observations. These values define the reported experiment; they are not asserted to be universally optimal.

7.3 Controlled simulation study

Simulations complement the financial study by providing known innovation laws. Stable multivariate VAR systems are generated under Gaussian, Student- t , mixture, and heteroskedastic regimes. Every innovation model is evaluated under every regime and across repeated random seeds. These experiments test recovery of known moments, quantiles, dependence, and forecast coverage; the financial case study evaluates external validity.

7.4 Ablation studies

Stub for thesis development. The ablation study will isolate the contribution of each CDI-VAR component using the same rolling origins, random seeds, VAR fits, and evaluation metrics. At minimum it will compare unconditional DI-VAR, residual-context conditioning without dynamic volatility, volatility conditioning without residual history, uncalibrated CDI-VAR, static regularized calibration, and the full adaptively calibrated CDI-VAR. The final section will state the estimands, experimental controls, paired uncertainty procedure, and results after the ablation experiments are frozen and run. No component-level causal claim is made before that analysis.

8 Hypotheses

1. Gaussian-VAR is competitive when residuals are approximately elliptical and stable.
2. Student- t -VAR improves tail calibration when heavy tails dominate but dependence remains relatively simple.
3. Bootstrap-VAR is competitive in low dimensions when sufficient representative residuals are available.
4. DI-VAR is most valuable when joint residual structure is non-elliptical or multimodal and the calibration sample supports flexible estimation.
5. CDI-VAR improves sharpness when recent standardized residuals and volatility contain information about the conditional innovation law.
6. Regularized adaptive calibration is more stable than unrestricted static calibration when forecast-error scale changes across regimes.
7. No unconditional innovation model remains calibrated under every conditional volatility or regime shift.
8. Dependence shifts can degrade portfolio-risk calibration even when marginal asset coverage remains acceptable.

9 Interpretation and Limitations

The framework is interpretable because it separates mean dynamics, residual uncertainty, calibration, and stress response. Nevertheless, VAR coefficients are predictive associations rather than causal effects. Financial returns may contain time-varying conditional means and variances that an unconditional residual model cannot represent. Rolling residual samples may be too small for stable diffusion training. Backtests reuse a finite and historically specific sequence of market conditions. Stress scenarios are sensitivity analyses rather than literal forecasts of future crises.

Most importantly, richer historical fit does not establish robustness. Diffusion may reproduce the reference residual distribution closely while extrapolating poorly under shift. Conclusions must therefore depend on out-of-sample calibration and controlled degradation, not visual sample quality alone.

10 Expected Contribution

The project contributes a common evaluation framework rather than a preferred algorithm:

- an interpretable decomposition of cross-asset forecasting and residual uncertainty;
- a leakage-aware rolling protocol for fitting joint innovation models;
- a fair comparison of parametric, empirical, and diffusion innovations;
- calibration diagnostics spanning marginal, joint, tail, and portfolio behavior;
- stress-response curves under volatility, dependence, tail, contamination, and Wasserstein perturbations;
- evidence describing when additional model complexity is warranted.

11 Future Research

The residual construction provides a direct path toward distributional-shift research. A monitoring system can compare recent residual windows with the reference distribution using Wasserstein distances, covariance and tail-dependence changes, two-sample statistics, or rolling calibration errors. Subsequent work can distinguish innovation shift from changes in VAR dynamics and determine whether to update the innovation model, refit the conditional mean, or widen robust uncertainty sets.

CDI-VAR implements one conditional law using recent standardized residuals and an EWMA volatility state. Future work can enlarge the information set with drawdown, liquidity, macroeconomic variables, or latent regimes, but each added state variable requires a nested evaluation design and an ablation establishing incremental value.

12 Conclusion

This study reframes multivariate financial forecasting as a sequence of distinct questions. A VAR asks what can be predicted from recent cross-asset behavior. A joint residual model asks what uncertainty remains. Calibration asks whether that uncertainty is credible. Stress testing asks how quickly credibility and financial risk estimates deteriorate when the innovation distribution changes.

Within this framework, DI-VAR is a flexible unconditional candidate and CDI-VAR is its volatility-aware, conditionally generated, explicitly calibrated extension. Comparing both with Gaussian, Student- t , and bootstrap alternatives makes the analysis interpretable, falsifiable, and useful even when simpler models prevail. The methodology connects probabilistic forecasting with distributional shift and robust risk evaluation while preserving clear boundaries between predictive dynamics, uncertainty representation, calibration, and stress response.

A Planned Outputs

The final study will include data and split summaries, VAR diagnostics, residual-distribution diagnostics, nominal calibration tables, proper scores, tail and portfolio metrics, computational comparisons, ablation results, and stress degradation summaries. Planned figures include return and volatility diagnostics, residual pair plots, PIT and reliability diagrams, forecast fan charts, joint downside diagnostics, portfolio loss distributions, and robustness curves over stress severity.

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